

Dataset Quality & Descriptive Statistics Report

Analysis and decision-support report for: ecommerce_final_clean_cleaned.csv

1. TENTANG DS GENERATOR

DS Generator (Descriptive Statistics Generator) adalah sebuah platform analisis data otomatis berbasis web yang dirancang khusus oleh **Kelompok 2 Data Science ITSB**. Aplikasi ini membantu HR, analis data, dan pengambil keputusan bisnis dalam mengunggah dataset, menganalisis kesehatan data (quality audit), membersihkan data yang kotor (data cleaning), menghitung statistik deskriptif tingkat lanjut, memvisualisasikan temuan secara interaktif, serta merumuskan rekomendasi keputusan strategis berbasis data secara real-time.

2. ANGGOTA TIM PENGEMBANG (KELOMPOK 2 ITSB)

Nama Lengkap	NIM	Peran / Fokus Analisis
Carol Dupino Pereira	52250051	Descriptive & Advanced Statistics Engine
Refantanur Husnul Haqib	52250052	Visualizations & Dynamic Plotly Dashboard
Cahaya Medina Semidang	52250053	Data Preprocessing & Sanitizer Module
Raihania Syah Putri	52250054	Time Series Forecasting & Trends Panel
Cloise Shafira	52250044	Smart Insights Generation Algorithm
Adinda Adelia Futri	52250055	Reporting System PDF/Excel & Security Sanitization

3. DESKRIPSI DATA & RINGKASAN EKSEKUTIF

Nama File Dataset:	ecommerce_final_clean_cleaned.csv
Waktu Analisis:	14 June 2026, 00:14:19
Status Kebersihan:	RAW DATA (Perlu Pembersihan)
Total Baris (Observasi):	1,576
Total Kolom (Variabel):	22
Variabel Numerik:	5
Variabel Kategorikal:	17

Missing Cells	Duplicate Rows	IQR Outliers
0 (0.0%)	0	8

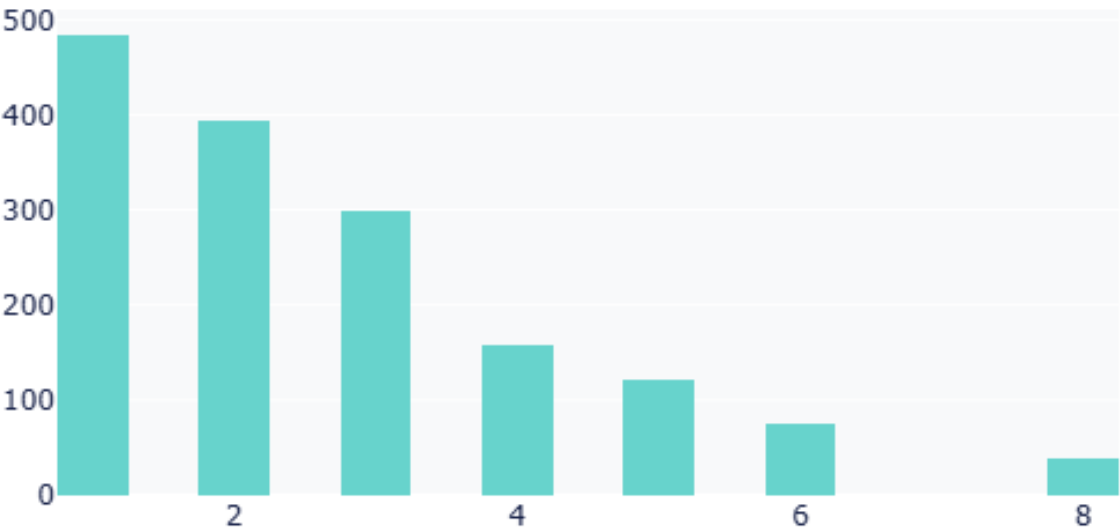
Identifikasi Masalah Kualitas Data:

- 8 outlier terdeteksi (metode IQR)

4. VISUALISASI DATA KUNCI

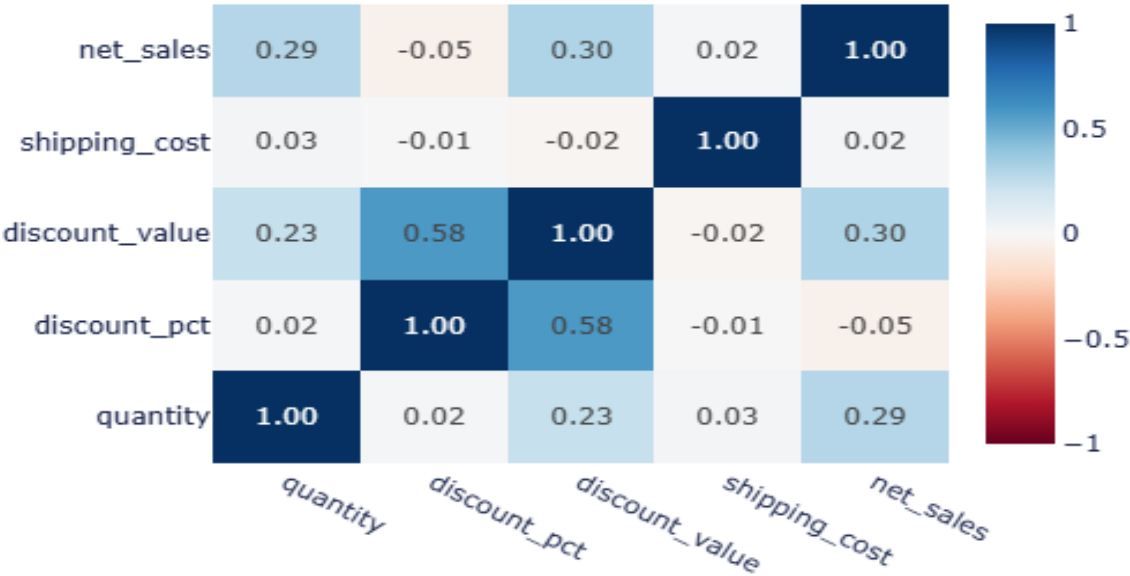
Histogram — quantity

Histogram: quantity



Correlation Heatmap

Correlation Heatmap



Bar Chart — order_date



5. TEMUAN KUNCI & INSIGHT OTOMATIS

Excellent Data Quality — Zero Missing Values

No missing values found at all. The dataset is clean and ready for statistical analysis or machine learning without additional preprocessing.

Dataset: 1,576 Rows × 22 Columns

Dataset size is sufficiently representative for reliable statistical analysis. Consists of 5 numeric and 17 categorical columns.

Highest Average Value: net_sales

Column **net_sales** has the highest average value of **746,658.62**. Median: 584676.5, Std: 637474.69.

Most Outliers: discount_value

Column **discount_value** has the most outliers: **73 data points (4.6%)** based on IQR method ($\pm 1.5 \times \text{IQR}$). Check whether these outliers are data errors or valid extreme values.

Highest Std Deviation: net_sales

Column **net_sales** has the largest standard deviation of **637,474.6918** (CV=85.4%). Very high variability on this column indicates a wide data spread.

Normality Test — 0/5 Columns Normal

NOT normal columns: quantity, discount_pct, discount_value (+2 more). Non-normal columns (100.0%) require non-parametric tests (Mann-Whitney, Kruskal-Wallis) or data transformation (log, sqrt, Box-Cox) before parametric analysis.

Skewed Distributions: 2 Column(s)

The following columns have skewness > 1: quantity (skew=1.13), discount_value (skew=3.59). These skewed distributions can affect parametric analysis results. Consider log or Box-Cox transformation.

Categorical Balance (order_date): Balanced

Category distribution is fairly even. Most frequent category: "01-15-2024" (0.4%). Total 949 unique categories in column order_date.

Further Analysis Recommendations

Based on the structure of this dataset, suggested techniques: One-Way ANOVA / T-Test to compare means across categorical groups; Linear / Logistic Regression for prediction and modeling; Chi-Square + Cramér's V for categorical variable association; Non-Parametric Tests (Mann-Whitney, Spearman) for non-normal columns; Principal Component Analysis (PCA) for dimensionality reduction.

6. REKOMENDASI STRATEGIS PENGAMBILAN KEPUTUSAN

R1. Investigasi Nilai Anomali (Outliers): Ditemukan beberapa data pencilan ekstrem. Disarankan tim operasional memvalidasi apakah angka tersebut valid karena fluktuasi riil bisnis atau human error, sebelum melakukan pemodelan prediktif lanjutan.

R2. Transformasi Skewness Data: Kolom (quantity, discount_value) memiliki kemiringan distribusi tinggi (skewed). Sebelum menerapkan algoritma berbasis asumsi distribusi normal, lakukan transformasi logaritma atau Box-Cox agar hasil prediksi memiliki bias minimal.

R3. Pembersihan Berkala (Data Governance): Jadwalkan proses pembersihan data secara rutin menggunakan DS Generator sebelum laporan akhir bulan diekspor ke departemen eksekutif, guna menjamin keputusan dibuat berdasarkan data yang higienis dan terpercaya.

7. DATA HEALTH & COLUMN DETAILS

Column Name	Type	Missing (%)	Unique	Outliers	Status / Issues
order_date	str	0 (0.0%)	949	0	OK
ship_date	str	0 (0.0%)	901	0	OK
platform	str	0 (0.0%)	5	0	OK
category	str	0 (0.0%)	6	0	OK
product_name	str	0 (0.0%)	25	0	OK
quantity	int64	0 (0.0%)	7	0	OK
campaign	str	0 (0.0%)	5	0	OK
voucher_code	str	0 (0.0%)	5	0	OK
discount_pct	float64	0 (0.0%)	5	0	OK
discount_value	str	0 (0.0%)	1044	0	OK
shipping_cost	int64	0 (0.0%)	7	0	OK
net_sales	int64	0 (0.0%)	1385	8	8 outliers
payment_method	str	0 (0.0%)	12	0	OK
customer_segment	str	0 (0.0%)	3	0	OK
region	str	0 (0.0%)	8	0	OK
stock_status	str	0 (0.0%)	3	0	OK
order_status	str	0 (0.0%)	8	0	OK
customer_rating	int64	0 (0.0%)	3	0	OK
priority_flag	str	0 (0.0%)	6	0	OK
is_high_value	str	0 (0.0%)	2	0	OK
order_priority	str	0 (0.0%)	3	0	OK
valid_transaction	str	0 (0.0%)	2	0	OK

8. STATISTIK DESKRIPTIF — VARIABEL NUMERIK

Column	Mean	Median	Min	Max	Std Dev	Mode	Skewness	Normality
quantity	2.66	2.0	1.0	8.0	1.69	1.0	1.1314	Not Normal
discount_pct	11.08	10.0	0.0	30.0	9.89	0.0	0.3864	Not Normal
discount_value	122349.83	61078.5	0.0	1994442.0	179698.69	0.0	3.5911	Not Normal
shipping_cost	14008.88	15000.0	0.0	25000.0	7645.97	20000.0	-0.4804	Not Normal
net_sales	746658.62	584676.5	0.0	2498967.0	637474.69	0.0	0.8745	Not Normal

9. STATISTIK DESKRIPTIF — VARIABEL KATEGORIKAL

Column	Unique	Mode (Most Freq)	Mode Freq	Mode %	Missing	Missing %
order_date	949	01-15-2024	6	0.38%	0	0.0%
ship_date	901	2024/03/31	187	11.87%	0	0.0%
platform	5	BLIBLI	326	20.69%	0	0.0%
category	6	FASHION	368	23.35%	0	0.0%
product_name	25	SNEAKERS	81	5.14%	0	0.0%
campaign	5	NORMAL DAY	519	32.93%	0	0.0%
voucher_code	5	NONE	543	34.45%	0	0.0%
payment_method	12	VIRTUAL ACCOUNT	320	20.3%	0	0.0%
customer_segment	3	RETURNING	857	54.38%	0	0.0%
region	8	BANDUNG	210	13.32%	0	0.0%
stock_status	3	IN STOCK	1154	73.22%	0	0.0%
order_status	8	COMPLETED	1283	81.41%	0	0.0%
customer_rating	3	5	1027	65.16%	0	0.0%
priority_flag	6	NO	524	33.25%	0	0.0%
is_high_value	2	NO	1117	70.88%	0	0.0%
order_priority	3	LOW	706	44.8%	0	0.0%
valid_transaction	2	VALID	1495	94.86%	0	0.0%

10. PERBANDINGAN METRIK SEBELUM & SESUDAH CLEANING

Metrik	Sebelum (Raw)	Sesudah (Cleaned)	Perubahan
Jumlah Baris	1576	1576	Dihapus: 0 baris
Jumlah Kolom	22	22	Dihapus: 0 kolom
Sel Kosong	0 (0.0%)	0 (0.0%)	Diisi/hapus: 0 sel
Baris Duplikat	0	0	Duplikasi 100% dihilangkan

11. LOG TINDAKAN PEMBERSIHAN DETIL PER KOLOM

Keterangan Modifikasi
• Shape: 1576×22 → 1576×22
• Whitespace stripped from text columns
• Empty strings converted to NaN